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 Studierichting: Informatica
 Oefeningenreeks 2B nr. 6.11

$$\begin{aligned}
 & 3 \operatorname{Bgtan} \frac{1}{4} + \operatorname{Bgtan} \frac{1}{20} + \operatorname{Bgtan} \frac{1}{1985} = \frac{\pi}{4} \\
 &= \operatorname{Bgtan} \frac{1}{4} + \operatorname{Bgtan} \frac{\frac{1}{4} + \frac{1}{4}}{1 - \frac{1}{4} \cdot \frac{1}{4}} + \operatorname{Bgtan} \frac{\frac{1}{20} + \frac{1}{1985}}{1 - \frac{1}{20} \cdot \frac{1}{1985}} \\
 &= \operatorname{Bgtan} \frac{1}{4} + \operatorname{Bgtan} \frac{8}{15} + \operatorname{Bgtan} \frac{2005}{39699} \\
 &= \operatorname{Bgtan} \frac{\frac{1}{4} + \frac{8}{15}}{1 - \frac{1}{4} \cdot \frac{8}{15}} + \operatorname{Bgtan} \frac{2005}{39699} \\
 &= \operatorname{Bgtan} \frac{47}{52} + \operatorname{Bgtan} \frac{2005}{39699} \\
 &= \operatorname{Bgtan} \frac{\frac{47}{52} + \frac{2005}{39699}}{1 - \frac{47}{52} \cdot \frac{2005}{39699}} \\
 &= \operatorname{Bgtan} \frac{\frac{1865853 + 104260}{2064348}}{\frac{1970113}{2064348}} \\
 &= \operatorname{Bgtan} \frac{2064348}{1970113} \\
 &= \operatorname{Bgtan} 1
 \end{aligned}$$

References